UNIVERSIDAD PERUANA DE CIENCIAS APLICADAS

FACULTY OF ENGINEERING

PROJECT REPORT - RECIPE RECOMMENDATIONS PROJECT

Subject: Big Data

NRC: 18517

Team members:

Full Name	**Student ID**
Marchan Quispe, Mildred Micaela	U202213292
Medina Agnini, Bruno Alessandro	U202216661
Miranda Rafael, Nicolás Francisco	U202216895
Rodríguez Valencia, Rosa Maria	U202212675

Professor: Alarcon Delgado Carlos Adrian

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1 Project Proposal

1.1 Domain

The domain of this project is **digital gastronomy and culinary recommendation systems**. In a context where millions of people search for recipes online daily, there is a critical need for intelligent systems that help users discover relevant recipes based on their preferences, dietary restrictions, available ingredients, and preparation patterns.

Food.com (formerly Genius Kitchen and Recipezaar) is one of the largest recipe platforms in the world, with over half a million recipes contributed by its community and millions of reviews accumulated over more than two decades.

1.2 Problem Statement

The main problem this project will address is:

> *How can we build an intelligent recipe discovery and recommendation system that leverages the underlying structure of ingredients, culinary techniques, and user preferences to suggest relevant recipes and discover emerging culinary patterns?*

Specific problems to solve:

1. **Discovery of similar recipes:** Identify recipes that share similar nutritional characteristics, ingredients, or preparation styles.
2. **Culinary segmentation:** Group recipes into coherent clusters representing culinary styles, dietary categories, or usage patterns.
3. **Personalized recommendation:** Suggest recipes based on the behavior of similar users and the intrinsic characteristics of the recipes.
4. **Ingredient graph analysis:** Discover relationships between ingredients, identify central ingredients in different culinary styles, and detect viable substitutions.
5. **Popularity and quality prediction:** Estimate which recipes will be well-received based on their characteristics and the publication context.

1.3 Expected Product Questions

The system should be able to answer questions such as:

- "Which recipes are similar to this pasta recipe I liked?"
- "What healthy recipes under 30 minutes can I prepare with chicken and broccoli?"
- "What are the most central ingredients in Mediterranean cuisine?"
- "What new recipes should I try based on my review history?"
- "What recipe clusters exist in my favorite desserts collection?"
- "What ingredients can I substitute if I don't have butter?"

1.4 Justification for Course Suitability

This dataset is particularly suitable for this course because it meets all structural requirements:

1. **Catalog Layer:** 522,517 recipes as main entities with rich metadata (name, author, category, preparation time, nutritional information, instructions).
2. **Features Layer:** Multiple types of features available:
   - Numerical: cooking times, nutritional information (11 metrics), servings.
   - Categorical: recipe categories, keywords.
   - Text: descriptions, instructions, ingredient lists.
   - Temporal: publication dates.
   - Structured lists: ingredients, quantities, instruction steps.
3. **Interaction Layer:** 1,401,982 reviews representing user-recipe interactions with explicit ratings (1-5 stars) and textual feedback.
4. **Graph Layer:** Multiple potential graphs:
   - Ingredient-ingredient graph (co-occurrence in recipes).
   - Recipe-recipe graph (based on ingredient similarity or users who rated them).
   - Author-author graph (collaboration in culinary styles).
   - Bipartite user-recipe projection.
5. **Pipeline Layer:** The project already has a reproducible and extensible ingestion script.

The project goes beyond basic supervised learning by integrating:

- Clustering to discover latent culinary segments.
- Dimensionality reduction for compact recipe representations.
- Recommendation systems (content-based, collaborative filtering, hybrids).
- Graph analysis to discover structure in ingredients and recipes.
- Complex feature engineering from text, lists, and nutritional data.

2 Data Sources Inventory

2.1 Main Source

Attribute	Detail
Name	Food.com Recipes and Reviews
Platform	Kaggle
URL	`https://www.kaggle.com/datasets/irkaal/foodcom-recipes-and-reviews`
Author/Curator	irkaal
Last update	2020 (static dataset)
License	CC0: Public Domain
Format	2 CSV files
Total size	∼450 MB (compressed)
Usability Score	10.0/10 (Kaggle)

Table 1: Data source information

2.2 Raw Files

File	Rows	Columns	Size
`recipes.csv`	522,517	28	∼350 MB
`reviews.csv`	1,401,982	8	∼100 MB

Table 2: Raw dataset files

2.3 Access Conditions

- **Public access:** The dataset is publicly available on Kaggle without requiring authentication.
- **Programmatic download:** Can be downloaded using the Kaggle API or the `kagglehub` library.
- **License:** CC0 (public domain) allows commercial and non-commercial use without restrictions.
- **Redistribution:** Allowed under the CC0 license.
- **Attribution:** Not legally required, but ethically recommended.

2.4 Data Provenance

The original data comes from **Food.com** (formerly Recipezaar, founded in 1999, later acquired and renamed as Genius Kitchen, and finally Food.com). The dataset was collected and curated by Kaggle user "irkaal" and published in 2020.

Time period: Recipes and reviews span from 1999 to 2020, providing over 20 years of culinary data.

3 Schema Draft

3.1 Entity Tables

3.1.1 Table: Recipes (Main Entity)

Field	Type	Description
RecipeId	int64	Unique recipe identifier (PK)
Name	object	Recipe name
AuthorId	int64	Author identifier (FK)
AuthorName	object	Author name
CookTime	object	Cook time (ISO 8601)
PrepTime	object	Prep time (ISO 8601)
TotalTime	object	Total time (ISO 8601)
DatePublished	object	Publication date
Description	object	Textual description
Images	object	Image URLs (list)
RecipeCategory	object	Recipe category
Keywords	object	Keywords (list)
RecipeIngredientQuantities	object	Quantities (list)
RecipeIngredientParts	object	Ingredients (list)
AggregatedRating	float64	Average aggregated rating
ReviewCount	float64	Number of reviews
Calories	float64	Calories per serving
FatContent	float64	Fat (g)
SaturatedFatContent	float64	Saturated fat (g)
CholesterolContent	float64	Cholesterol (mg)
SodiumContent	float64	Sodium (mg)
CarbohydrateContent	float64	Carbohydrates (g)
FiberContent	float64	Fiber (g)
SugarContent	float64	Sugar (g)
ProteinContent	float64	Protein (g)
RecipeServings	float64	Number of servings
RecipeYield	object	Textual yield
RecipeInstructions	object	Instruction steps (list)

Table 3: Recipes table schema

3.1.2 Table: Reviews (Interactions)

Field	Type	Description
ReviewId	int64	Unique review identifier (PK)
RecipeId	int64	Recipe identifier (FK)
AuthorId	int64	Reviewer identifier (FK)
AuthorName	object	Reviewer name
Rating	int64	Rating (1-5 stars)
Review	object	Review text
DateSubmitted	object	Submission date
DateModified	object	Modification date

Table 4: Reviews table schema

3.2 Keys and Relationships

- **Primary Keys:**
  - `recipes.RecipeId`
  - `reviews.ReviewId`
- **Foreign Keys:**
  - `reviews.RecipeId` → `recipes.RecipeId`
  - `recipes.AuthorId` (Recipe author)
  - `reviews.AuthorId` (Review author)
- **Relationships:**
  - Recipes ↔ Reviews: One-to-Many (1:N)
  - Recipes ↔ Authors: Many-to-One (N:1)
  - Reviews ↔ Reviewers: Many-to-One (N:1)

3.3 Expected Joins

1. **Recipe-Review Join:**

```sql
SELECT r.*, rev.Rating, rev.Review, rev.DateSubmitted
FROM recipes r
LEFT JOIN reviews rev ON r.RecipeId = rev.RecipeId
```

Listing 1: Main join for interaction analysis

2. **Recipe Aggregations:**

```sql
SELECT
    RecipeId,
    COUNT(*) as num_reviews,
    AVG(Rating) as avg_rating,
    STDDEV(Rating) as rating_variance
FROM reviews
GROUP BY RecipeId
```

Listing 2: Review metric aggregations

3. **Author Analysis:**

```sql
SELECT
    AuthorId,
    AuthorName,
    COUNT(DISTINCT RecipeId) as num_recipes,
    AVG(AggregatedRating) as avg_recipe_rating
FROM recipes
GROUP BY AuthorId, AuthorName
```

Listing 3: Author-based analysis

4 Processed Dataset V1

4.1 Ingestion Process

The ingestion process is implemented in the `ingest_foodcom_data.py` script, which performs the following transformations:

4.1.1 Transformation Pipeline

1. **Automatic download:** Using `kagglehub` to download the dataset from Kaggle.
2. **List parsing:** Columns stored as strings are parsed into Python lists:
   - `Images`
   - `Keywords`
   - `RecipeIngredientQuantities`
   - `RecipeIngredientParts`
   - `RecipeInstructions`
3. **Duration parsing:** Conversion from ISO 8601 format to minutes:
   - `CookTime` → `CookTime_Minutes`
   - `PrepTime` → `PrepTime_Minutes`
   - `TotalTime` → `TotalTime_Minutes`
4. **Initial feature engineering:**
   - `NumIngredients`: Count of ingredients.
   - `NumQuantities`: Count of specified quantities.
5. **Name normalization:** Conversion to snake_case for consistency.
6. **Persistence:** Saved in processed CSV format.

4.1.2 Execution Command

The script is fully reproducible and executed with:

```
# Dependency installation
pip install pandas isodate kagglehub

# Script execution
python ingest_foodcom_data.py --output-dir ./data/processed
```

Listing 4: Ingestion command

4.2 Generated Processed Files

File	Location	Description
`recipes_processed.csv`	`data/processed/`	Cleaned and enriched recipes
`reviews_processed.csv`	`data/processed/`	Normalized reviews

Table 5: Processed dataset V1 files

4.3 Pipeline Validation

To verify the correct execution of the pipeline:

```python
import pandas as pd

# Load processed data
recipes = pd.read_csv('data/processed/recipes_processed.csv')
reviews = pd.read_csv('data/processed/reviews_processed.csv')

# Basic validations
assert len(recipes) == 522517, "Incorrect number of recipes"
assert len(reviews) == 1401982, "Incorrect number of reviews"

# Validate that time columns were created
assert 'CookTime_Minutes' in recipes.columns
assert 'PrepTime_Minutes' in recipes.columns
assert 'TotalTime_Minutes' in recipes.columns

# Validate derived features
assert 'NumIngredients' in recipes.columns
assert recipes['NumIngredients'].notna().sum() > 0

print("    Pipeline successfully validated")
```

Listing 5: Validation script

5 Data Dictionary Draft

5.1 Table: Recipes Processed

Field	Type	Description	Values/Range
RecipeId	int64	Unique recipe ID	Positive
Name	string	Recipe name	Free text
AuthorId	int64	Author ID	Positive
AuthorName	string	Author name	Free text
CookTime	string	ISO Cook time	e.g.: PT30M
PrepTime	string	ISO Prep time	e.g.: PT15M
TotalTime	string	ISO Total time	e.g.: PT45M
CookTime_Minutes	float64	Cooking in minutes	0.0 - 10000+
PrepTime_Minutes	float64	Prep. in minutes	0.0 - 5000+
TotalTime_Minutes	float64	Total in minutes	0.0 - 15000+
DatePublished	string	Publication date	YYYY-MM-DD
Description	string	Description	Free text
Images	list	Image URLs	List of strings
RecipeCategory	string	Category	Desserts, Main, etc.
Keywords	list	Keywords	List of strings
RecipeIngredient- Quantities	list	Quantities	List of strings
RecipeIngredient- Parts	list	Ingredients	List of strings
NumIngredients	int64	No. of ingredients	1 - 50+
NumQuantities	int64	No. of quantities	0 - 50+
AggregatedRating	float64	Average rating	0.0 - 5.0
ReviewCount	float64	Total reviews	0 - 10000+
Calories	float64	Calories/serving	0.0 - 5000+
FatContent	float64	Fat (g)	0.0 - 500+
SaturatedFatContent	float64	Sat. fat (g)	0.0 - 200+
CholesterolContent	float64	Cholesterol (mg)	0.0 - 1000+
SodiumContent	float64	Sodium (mg)	0.0 - 10000+
CarbohydrateContent	float64	Carbohydrates (g)	0.0 - 500+
FiberContent	float64	Fiber (g)	0.0 - 100+
SugarContent	float64	Sugar (g)	0.0 - 300+
ProteinContent	float64	Protein (g)	0.0 - 300+
RecipeServings	float64	Servings	1 - 100+
RecipeYield	string	Textual yield	Free text
RecipeInstructions	list	Steps	List of strings

Table 6: Data dictionary - Recipes Processed

5.2 Table: Reviews Processed

Field	Type	Description	Values/Range
ReviewId	int64	Unique review ID	Positive
RecipeId	int64	Recipe ID (FK)	Positive
AuthorId	int64	Reviewer ID	Positive
AuthorName	string	Reviewer name	Free text
Rating	int64	Rating	1, 2, 3, 4, 5
Review	string	Review text	Free text or NULL
DateSubmitted	string	Submission date	YYYY-MM-DD
DateModified	string	Modification date	YYYY-MM-DD

Table 7: Data dictionary - Reviews Processed

5.3 Notes on Data Types

- **Lists (list):** In the processed CSV, lists are stored as strings containing Python representations of lists. Using `ast.literal_eval` during loading converts them to actual lists.
- **Dates:** Stored as strings in ISO 8601 format. Conversion to `datetime` is required for temporal analysis.
- **ISO 8601 Durations:** Standard format like `PT30M` (30 minutes). The `*_Minutes` columns are the numerical version.
- **Null values:** Represented as `NaN` in numerical columns and empty strings or NULL in text columns.

6 Scale Analysis

6.1 Dataset Dimensions

Metric	Recipes	Reviews
Number of rows	522,517	1,401,982
Number of columns (raw)	28	8
Number of columns (processed)	31	8
In-memory size (MB)	∼111.6	∼85.6
Disk CSV size (MB)	∼350	∼100

Table 8: Dataset dimensions

6.2 Missingness Analysis

6.2.1 Recipes Table

Field	Non-Null	Null Count	% Missing
RecipeId	522,517	0	0.00%
Name	522,517	0	0.00%
AuthorId	522,517	0	0.00%
AuthorName	522,517	0	0.00%
CookTime	439,972	82,545	15.80%
PrepTime	522,517	0	0.00%
TotalTime	522,517	0	0.00%
DatePublished	522,517	0	0.00%
Description	522,512	5	0.00%
Images	522,516	1	0.00%
RecipeCategory	521,766	751	0.14%
Keywords	505,280	17,237	3.30%
AggregatedRating	269,294	253,223	48.46%
ReviewCount	275,028	247,489	47.37%
Nutritional Data	522,517	0	0.00%

Table 9: Missingness analysis - Recipes

Key observations:

- **CookTime (15.80% missing):** Some recipes do not specify cook time (e.g., salads).
- **AggregatedRating and ReviewCount (∼48% missing):** Almost half of the recipes do not have reviews yet.
- **Nutritional data (0% missing):** Critical fields are entirely complete.

6.3 Sparsity and Memory Estimations

6.3.1 User-Item Matrix (Reviews)

- **Unique Users:** Estimated ∼270,000.
- **Items (Recipes):** 522,517.
- **Interactions:** 1,401,982.
- **Density:** $\approx 0.00099\%$ (extremely sparse).

Implications:

- Requires sparse matrix techniques (scipy.sparse, CSR format).
- Matrix Factorization is ideal for this level of sparsity.

7 Ethics and Access Note

7.1 Data Origin

Data for this project comes from **Food.com**, a public platform. The dataset was curated and published on Kaggle under the **CC0: Public Domain** license.

7.2 Ethical Risk Mitigation

- **Pseudonymization:** User IDs are used instead of real names where possible.
- **Public Domain:** The data is already public and provided voluntarily by users.
- **No Sensitive Data:** No financial or health data is included.

8 Week 3 Conclusions

8.1 Achievements

The ingestion pipeline is functional and the data schema is defined. Scale analysis confirms that sparse matrices are required for processing interactions.